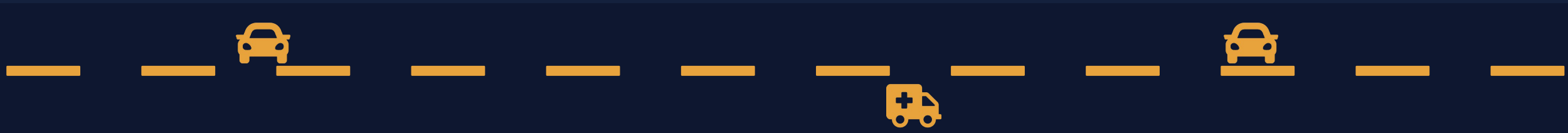
THEME 2 · AI-POWERED TRAFFIC OPTIMISATION FOR CHANDIGARH

Event Traffic Rehearsal

Test a big event's traffic plan in a simulation of Chandigarh's roads, before the crowd arrives.

Gemma 4 runs offline as the planner.
SUMO simulates every plan.
An officer approves every decision.



Krishna · Code for Communities Chandigarh, GDG Cloud Chandigarh

THE PROBLEM, FROM DECEMBER 2024

Two concerts at Sector 34. The city learnt on the day.

20,000+

fans at the Karan Aujla show, 7 Dec 2024

11 AM

road to the ground closed on 14 Dec, for an evening show

5 hospitals

near the ground; injured fans struggled to get help

Traders + doctors

asked the city to move the next shows

- **7 Dec** — **Karan Aujla, Sector 34**
  Jams at Piccadilly and New Labour Chowk, arriving and leaving
- **13 Dec** — **City decides**
  No more huge gatherings at Sector 34 in future
- **14 Dec** — **Diljit Dosanjh, Sector 34**
  5 parking areas, shuttles, 3 diversions. Still cars parked on roads
- **21 Dec** — **AP Dhillon moved to Sector 25**
  The new venue also caused inconvenience
- **Aug 2025** — **Sarangpur venue proposed**
  35,000 people, 6,250 parking spaces. No traffic plan described

No plan was tested before the day. Every fix was trial and error on real roads.

OUR SOLUTION

A rehearsal loop: the agent proposes, the simulator tests

1 Describe

Officer pastes the event notice. Gemma fills in venue, times, crowd, parking

2 Check

Officer corrects the details and sees every place on the map

3 Rehearse

SUMO plays the evening on Chandigarh's roads, 3 times per plan

4 Compare

Plans side by side: delay, ambulances, clearing time, map

5 Approve

Approver signs off. Advisory in English, Hindi, Punjabi

Round 1 is the officer's own plan.

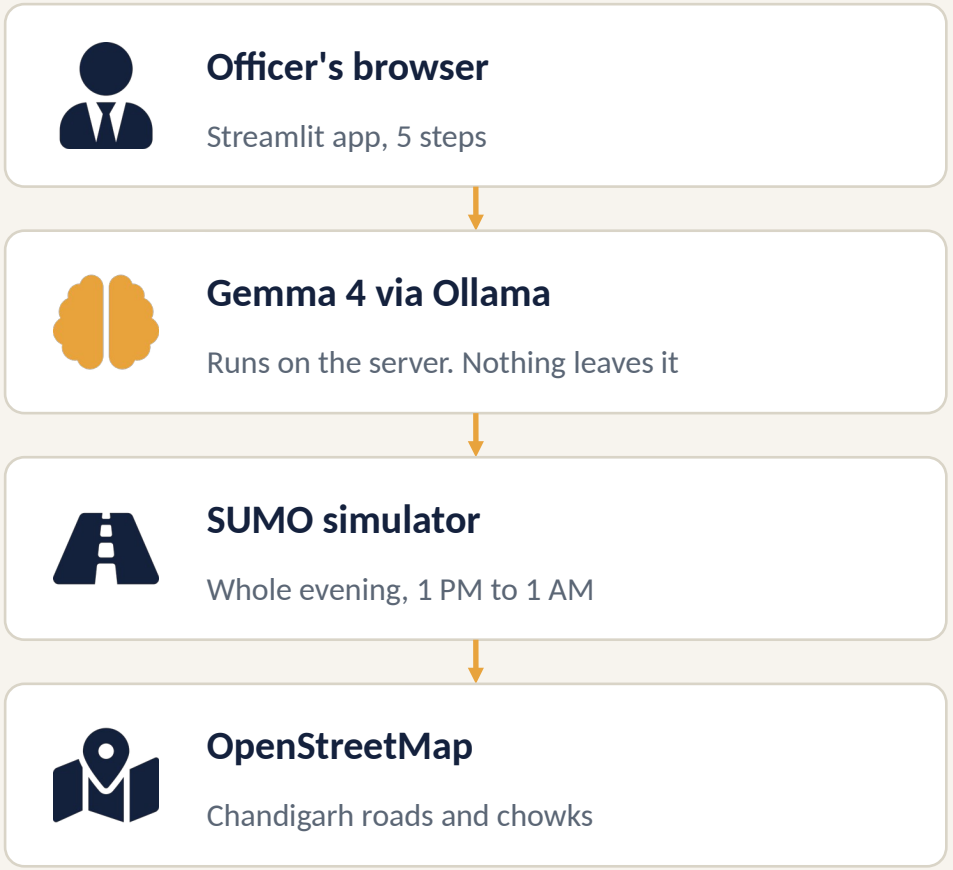
In our demo that is the real police plan for 14 Dec 2024: closures, 5 parking areas, shuttles, 3 diversions.

Up to 5 rounds

Gemma reads the results, names the worst bottleneck, makes up to 3 changes, and the plan is tested again. Worse plans are discarded.

UNDER THE HOOD

Open models, real roads, and rules the AI can't break



TWO HARD RULES

- Ambulances to protected hospitals: at most 5 extra minutes
- No gridlock: a plan that jams more than 1% of vehicles is rejected

Guardrails that keep a small model reliable

- Fixed JSON answers, checked; errors go back to Gemma (2 retries)
- Only menu actions and named places; typos like "Picadilly" are matched
- At most 3 changes per round, so it is clear what helped
- Every number Gemma writes is checked against the results
- Round, run and time limits enforced in code, not by the model

WHAT THE OFFICER SEES

Every plan judged on the same numbers

Total extra delay

vehicle-hours, for visitors and everyone else

Ambulance check

pass or fail per hospital, in minutes

Time to clear

minutes after the show until roads are normal

Market closures

hours of closures in shops and homes

Parking

cars per lot, when each lot fills up

Walk from parking

average minutes, with shuttles counted

Map: roads coloured by extra wait, with minutes written out for colour-blind users

< 1 min 1 to 3 3 to 10 > 10 min

QUESTIONS THE DEMO ANSWERS

Could 14 Dec have gone better?

The real police plan vs the agent's plans

Was Sector 25 really better?

Same crowd, the plan used for AP Dhillon

"What if we close X?"

The agent tests it and says so honestly, even if it's worse

CONCLUSION

Rehearse the next big event before the day.

Private by design

One government server. Gemma runs offline; no event or VIP data leaves it

Open and free to run

Apache-2.0 code, open-weight Gemma, SUMO, OpenStreetMap. No training costs

Fits how the city works

Complements the SOP for large gatherings the administration began drafting in Dec 2024

At the finale we build next

Rehearse the new Sarangpur venue before it is built · rallies and VIP convoys as moving closures · several events on one day · UX4G front end with logins and audit trail

github.com/<your-account>/event-traffic-rehearsal

Map © OpenStreetMap contributors · SUMO (EPL-2.0) · Gemma 4 (Apache 2.0) · Not an official Chandigarh Police product